Amazon EFS

Practical Task

Shared Storage Between Two EC2 Instances

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Task 1: What is Amazon EFS?

Amazon Elastic File System (EFS) is a fully managed, serverless, elastic file storage service from AWS. It exposes a standard NFS file system that many EC2 instances can mount and use at the same time, and it grows and shrinks automatically as files are added or removed, so no capacity planning is needed.

Key features

• Shared access: thousands of instances can mount the same file system concurrently (NFSv4).
• Elastic: scales automatically up to petabytes; pay only for storage used.
• Highly available and durable: data is stored redundantly across multiple Availability Zones.
• Secure: encryption at rest and in transit, IAM and security-group based access control.
• Storage classes (Standard, Infrequent Access, Archive) with lifecycle management to cut cost.

Common use cases

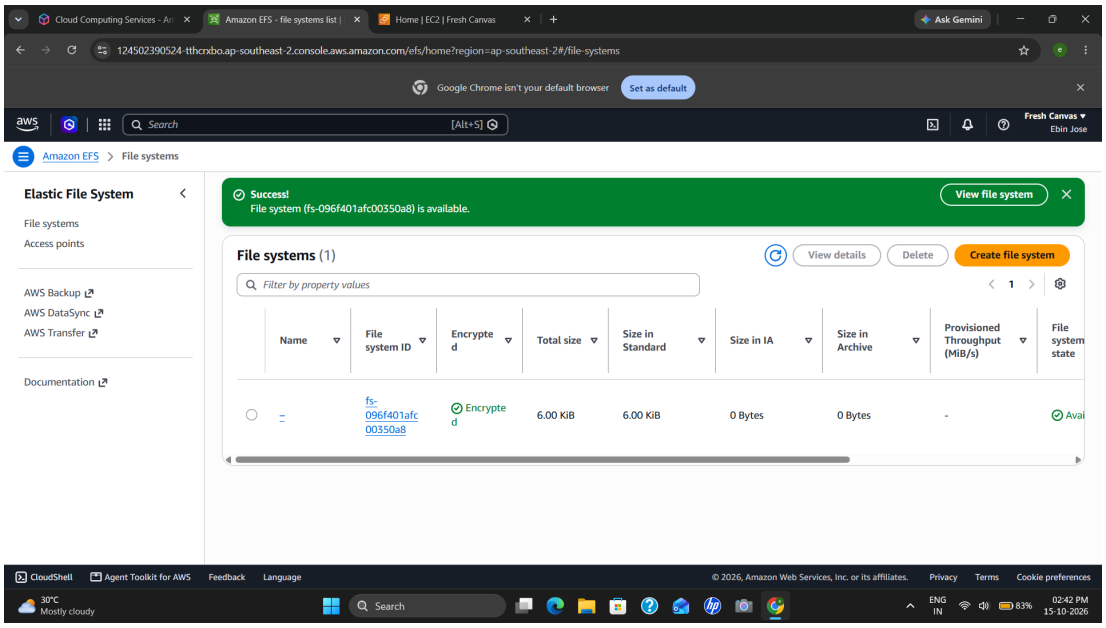
• Content management, web serving and shared website files.
• Shared home directories and application data across a fleet of servers.
• Big data analytics, machine learning and media processing.
• Containers and serverless workloads (ECS, EKS, Lambda) needing persistent shared storage.

EFS vs EBS

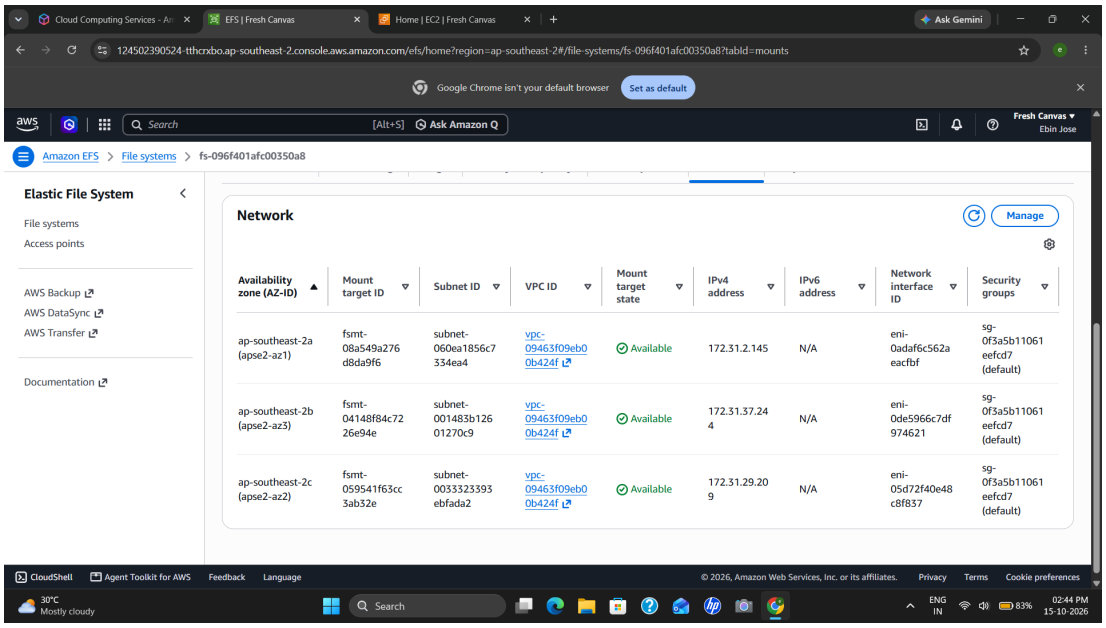
• EFS is shared network file storage (NFS); EBS is block storage attached like a virtual disk.
• EFS can be mounted by many instances across AZs at once; EBS normally attaches to one instance in one AZ.
• EFS scales automatically; EBS has a fixed size that is resized manually.
• EFS is best for shared files and web content; EBS is best for boot volumes and databases.

Task 2: Create an Amazon EFS File System

An encrypted EFS file system was created in the default VPC. The file system shows state **Available**, and mount targets exist in all three Availability Zones with state **Available**.



EFS > File systems list showing fs-096f401afc00350a8 available

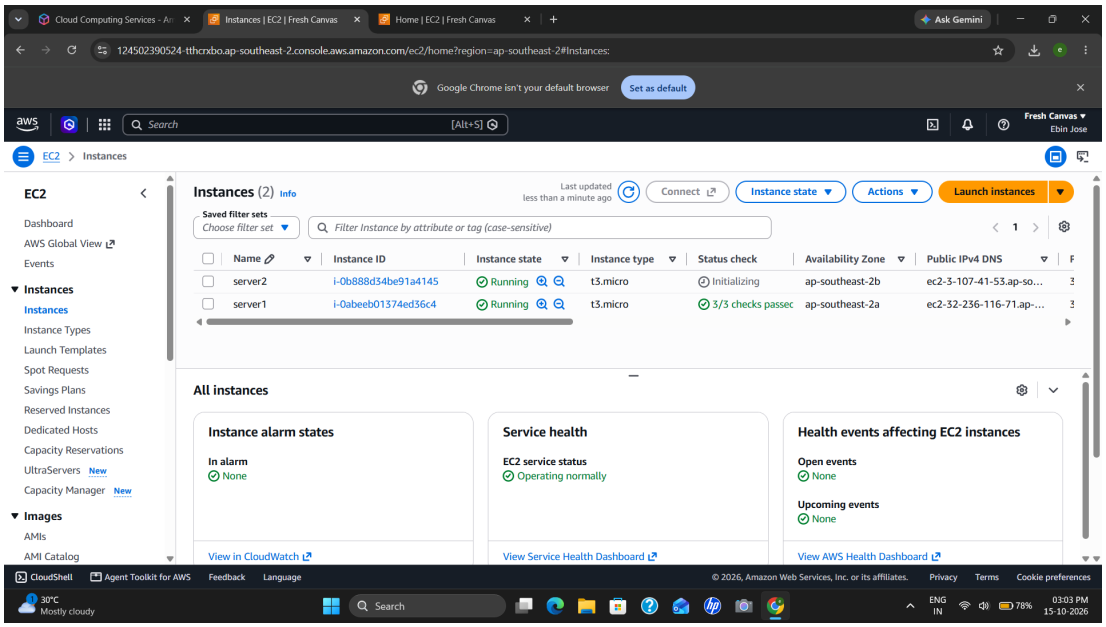


EFS > Network tab showing three mount targets in state Available

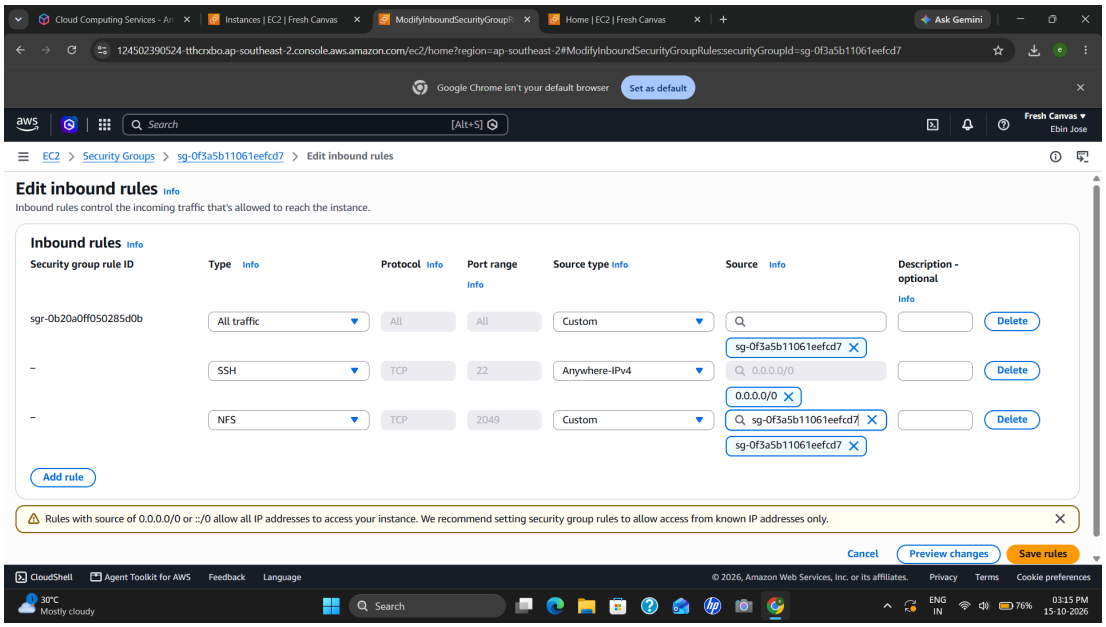
Task 3: Launch Two EC2 Instances

Two Amazon Linux 2023 (t3.micro) instances were launched in the same VPC as the file system, in different Availability Zones. Both are in the Running state.

Security group inbound rules allow SSH (22) and NFS (2049).



EC2 > Instances list showing server1 and server2 running



Security group > Edit inbound rules showing SSH (22) and NFS (2049)

Task 4: Mount the EFS

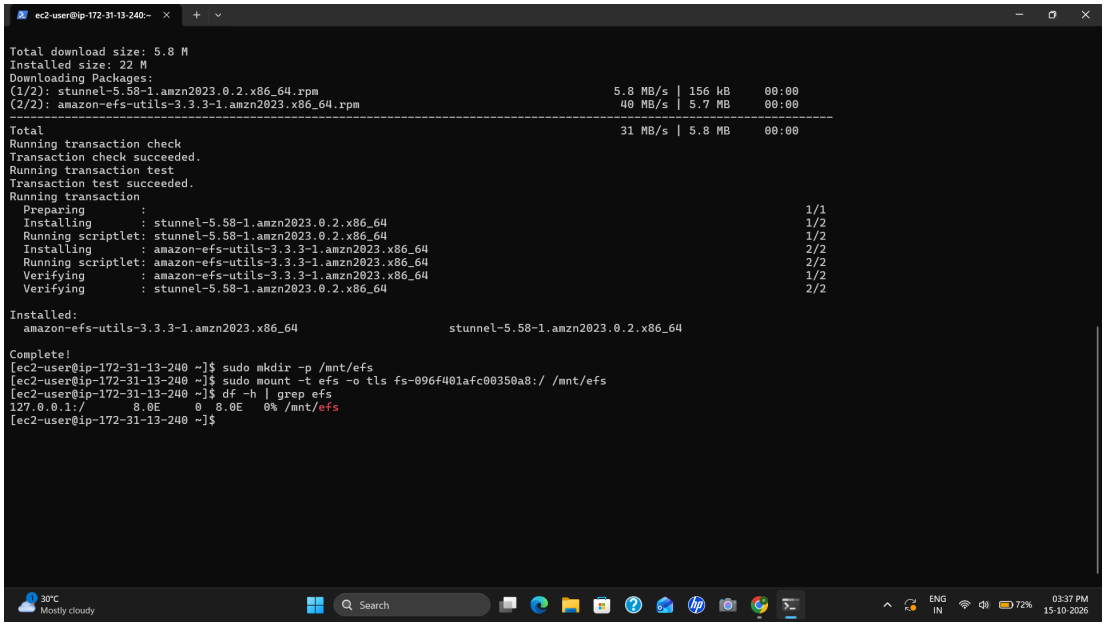
The same steps were run on both servers: install the EFS mount helper, create a mount point, mount the file system with TLS, and verify with `df -h`.

```
sudo dnf install -y amazon-efs-utils
sudo mkdir -p /mnt/efs
sudo mount -t efs -o tls fs-096f401afc00350a8:/ /mnt/efs
df -h | grep efs
```

Output on both servers: the file system is mounted at /mnt/efs, showing 8.0E size and 0% used.

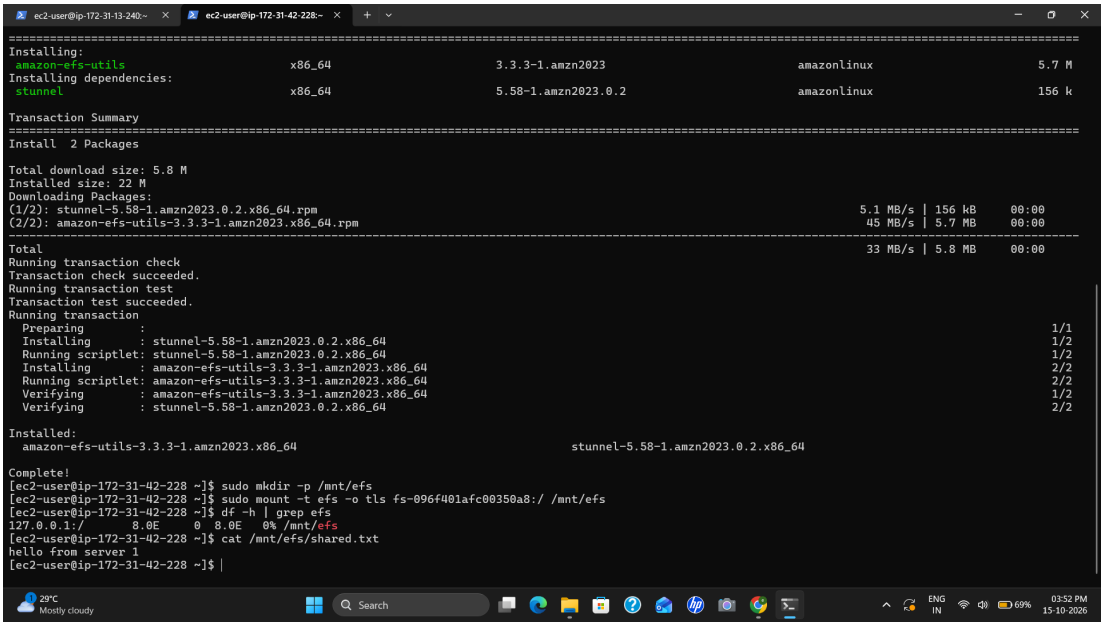
```
127.0.0.1:/     8.0E     0     8.0E     0%   /mnt/efs
```

Server 1 (ip-172-31-13-240)



Server 1 terminal: install, mount and df -h output

Server 2 (ip-172-31-42-228)

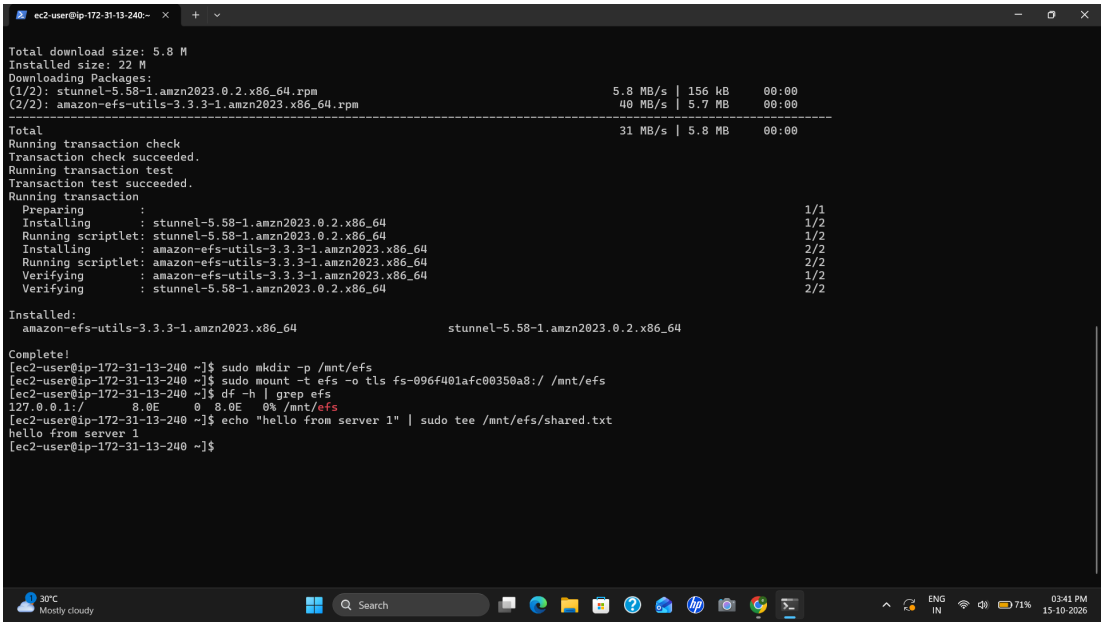


Server 2 terminal: install, mount and df -h output

Task 5: Verify Shared Storage

Step 1 - Create a file on Server 1

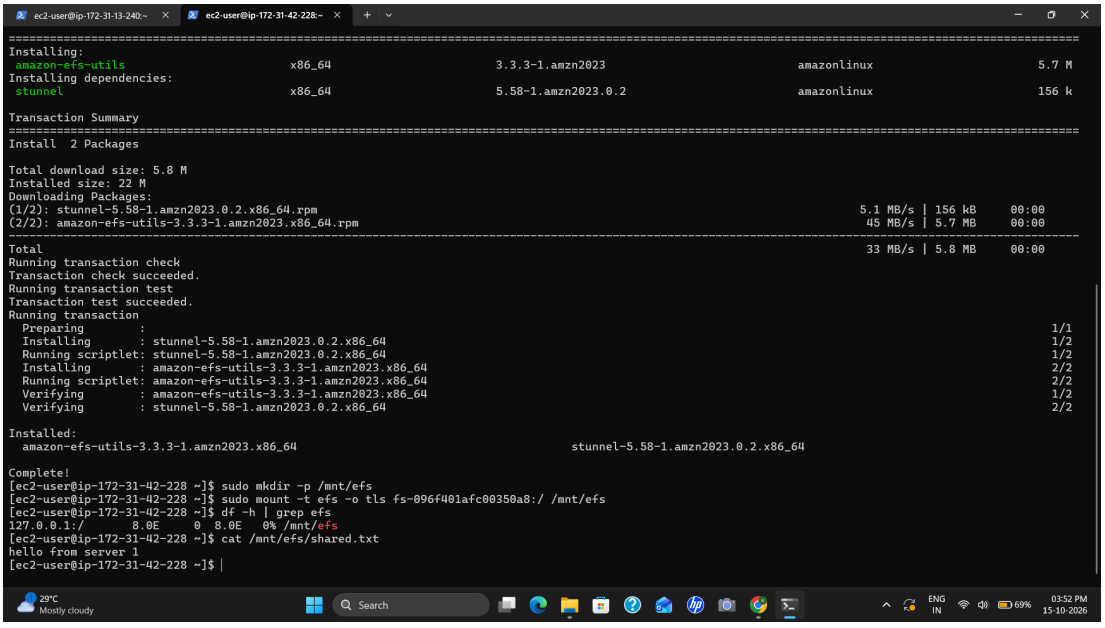
```
echo "hello from server 1" | sudo tee /mnt/efs/shared.txt
```



Server 1: shared.txt created

Step 2 - Read the file on Server 2

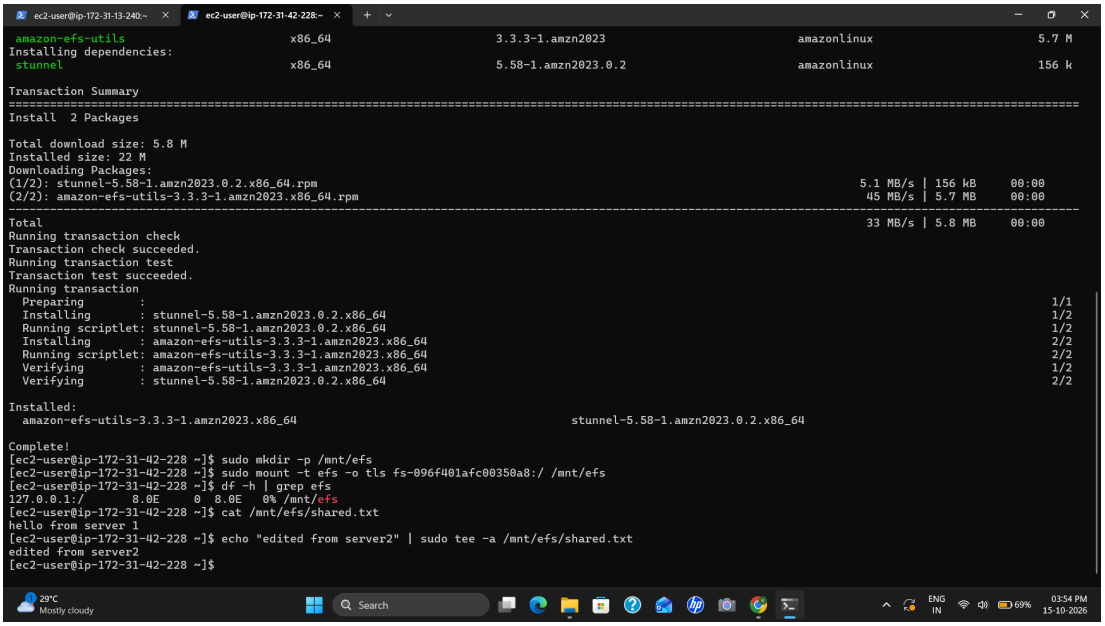
```
cat /mnt/efs/shared.txt
# hello from server 1
```



Server 2: shared.txt visible with the same content

Step 3 - Edit the file from Server 2

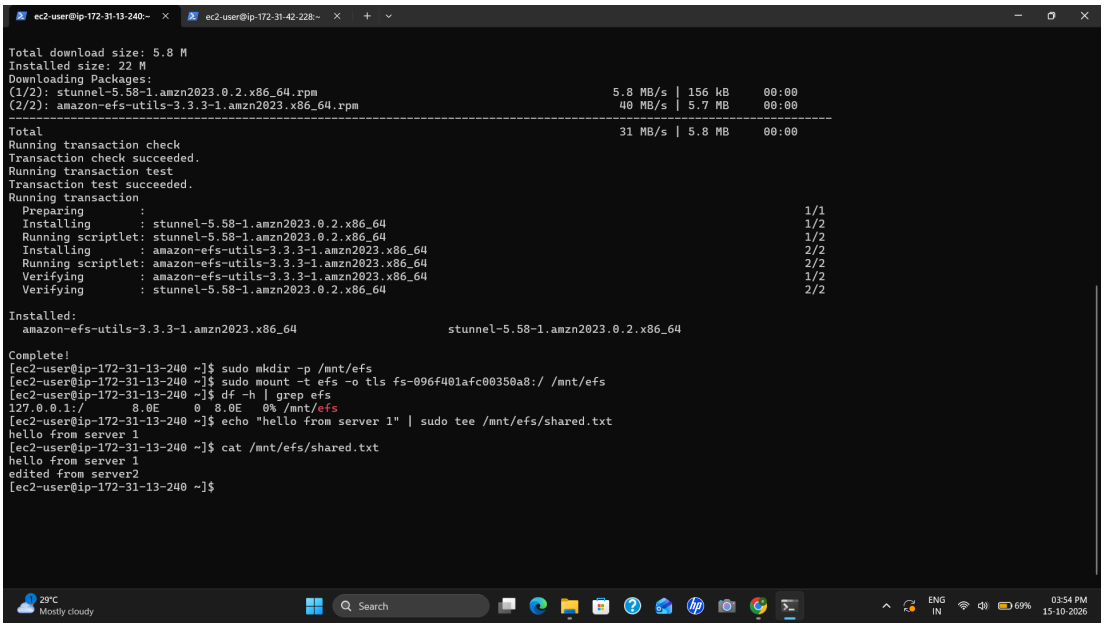
```
echo "edited from server2" | sudo tee -a /mnt/efs/shared.txt
```



Server 2: line appended to shared.txt

Step 4 - Confirm the change on Server 1

```
cat /mnt/efs/shared.txt
# hello from server 1
# edited from server2
```



Server 1: file now shows the line added from Server 2